

-----Original Message-----

**From:** Ulrich Poeschl [SMTP:ulrich.poeschl@ch.tum.de]  
**Sent:** Thursday, July 27, 2000 8:55 AM  
**To:** Seeman, Jeffrey I  
**Subject:** Re: draft paper

2069750139

Dear Jeff,

up to now I have the impression that the basic approach for the kinetic model is o.k., but I have not had the time to carefully go through the details and question every single step (implying that it appears to be correct at first sight but that I would not yet bet on it).

One concern I have about the relevance of the model results with respect to smoking is the neglect of ammonia re-absorption. I think that a more detailed discussion of this aspect is needed, providing examples in absolute numbers which demonstrate that evaporation can really be considered independently of the reverse effect of re-absorption/condensation. Reabsorption/condensation will always occur at least to some extent. I think in order to judge whether it is really negligible one has to consider diffusion in the gas and liquid phase and the absolute time scale in addition to the mere ratio of gas phase to particle volume.

Also I wonder if relative rates are really sufficient to judge the effect of ammonia in actual tobacco smoke. If the real residence time of the smoke would correspond to the first three "relative" seconds of your simulation only, then the effect of ammonia might be very significant since it takes these three "relative seconds" to drive the excess ammonia out of the particles.

Another question: Is it justified to assume that at the beginning ammonia has no significant gas phase concentration and how does the amount of ammonia you assume to be in the smoke particles (volume times conc.) compare to the total amount of ammonia available in the tobacco/tobacco smoke?

Best regards,  
Uli

P.S.: The typewriter font you used is a little hard to read. Maybe also the referees would appreciate a Times New Roman font.

At 13:44 26.07.00 -0400, you wrote:

>Dear colleagues,

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>I am currently planning to visit Emory University early next week to discuss  
>the status of the revised paper, the one that I left with you for your  
>review and input. It would be wonderful if I could have your ideas before